Modeling the disaster relief hub network design problem: A case of Central Vietnam

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Abstract. Disaster in Vietnam (typhoons, floods, ...) is severe and cause many damage to people and assets. An important phase in search and rescue (SAR) operations is to establish a reliable and optimal humanitarian logistics network. However, there is very little previous work addressed this problem of establishing an optimal network structure considering the harsh environmental constraints of reality (there maybe many impassable roads, or the movement costs could be very high and the routes are not stable), especially in the context of Vietnam. We aim to help the situation by contribute useful solutions. First, we modeled the problem as an multi-objective incomplete hub network design problem. Next, to develop an efficient MOEA algorithm, we define corresponding encoding and decoding scheme and integrate it into the NSGA-II framework. The performance of the algorithm is assessed extensively through the standard benchmark datasets TR, AP and CAB. The results show that the proposed algorithm statistically outperforms standard baselines. The solutions found by our algorithm on the Vietnam case study are then analyzed to interpret its implication for decision makers.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [2, 3]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [7, 9].

There were many great challenges in humanitarian operations during this period, one of them being the severe disruption of national logistics infrastructure. The North-South Railway – backbone of Vietnam's transport network was paralyzed due to track erosion and submersion, forcing the cancellation of 170 trains [7]. Simultaneously, key arteries such as the National Highway 1 and routes in the Central Highlands were severed by landslides, isolating communities for days [5]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed. The physical network becomes "incomplete", characterized by broken links and isolated nodes. Another challenge is resource scarcity resulting from poor planning and uncertain supply and demand. Disaster management involves four phases: mitigation, preparedness, response, and recovery [1]. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related works – AI-generated, has not re-written

Related Works (Current effort to resolve the disaster management problem (DRND))

Research Gaps

To bridge these gaps, this study proposes a novel approach with the following contributions:

1. We develop a **Multi-modal Hub-and-Spoke network model** that explicitly considers infrastructure disruption, resulting in an incomplete hub network structure that reflects realistic disaster conditions.
2. We formulate a **Robust multi-objective model** utilizing interval uncertainty to handle the unpredictability of demand and supply parameters effectively.
3. We apply efficient **MOEAs, specifically NSGA-II**, to solve the proposed model, demonstrating its practical applicability through a case study of the Central Vietnamese Floods.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND), specifically adjusted for flood disaster response in Central Vietnam, we framed the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP). This model is the capacitated, single-allocation variant of the HLP.

The main components of our network are origins, hubs, and demands. In our practical settings, hubs refer to the rescue stations. There are two kinds of hubs, corresponding to two kinds of strategies. The first type is established in advance of the disasters, strategically positioned, and inventoried with amenities. The second type is opened dynamically for disaster response and has high approachability to affected areas. Lastly, during the disaster response, we gradually discover the victim groups, which are referred to as **demands**. Besides the government's planned forces, some volunteers spontaneously offer to help; these are referred to as **origins**.

From a practical point of view, DRND involves multiple critical uncertainty factors. Firstly, the disruption of transportation infrastructure (e.g., damaged roads and buildings) can easily disable pre-programmed plans, urging the need for a dynamic solution [10]. Secondly, the exact locations of the demands, the total number of victims, and the availability of supply origins are highly uncertain during the rescue process [8].

To account for these uncertainties, we employ a two-stage stochastic programming approach utilizing multiple scenarios. In each scenario, a specific risk factor is assigned to each area. We also assume that search and rescue (SAR) operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy the uncertain demands under extreme scenarios.

Finally, the model places a strong emphasis on *social equity* by integrating the deprivation cost [6]. To obtain a robust approximation of the rescue time from an established hub to its assigned demand points without causing computational intractability, we utilize Daganzo's continuous approximation formula [4], a technique widely adopted in routing problems.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned} \text{Min } Z_1 = & \underbrace{\sum_{k\in\mathcal{H}} F_k x_k + \sum_{k\in\mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\ & + \sum_{s\in\mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k\in\mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j\in\mathcal{J}} \sum_{k\in\mathcal{H}} \left(\min_{m\in\mathcal{M}|a_{jkms}=1} C_{jkm} \right) O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\ & + \underbrace{\sum_{k\in\mathcal{H}} \sum_{h\in\mathcal{H}} \sum_{m\in\mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k\in\mathcal{H}} \sum_{i\in\mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg] \end{aligned} \quad (1)$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [4]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost Θ_{kis} is pre-computed as a parameter:

$$\Theta_{kis} = \min_{m \in \mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is} \cdot \Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [6], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k \in \mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned} &\text{Minimize} \quad Z_1, Z_2 \\ &\text{subject to} \quad \text{Constraints (3)} - (16) \end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{3}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{4}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{7}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \tag{12}$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} \leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} & \\ + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} & \end{aligned} \tag{13}$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{15}$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \tag{16}$$

Constraints (3) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (4), while the reactive hubs are required to be placed within safe zones (5). Constraints (6)-(11) rigorously enforce the single-allocation property to active hubs via available links. Constraint (12) restricts flow to be trans-shipped on intact paths. Constraints (13) and (14) maintain flow conservation and capacity limits. Specifically, (13) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (15)-(16).

3 Solution approach (Algorithms)

3.1 NSGA-II Framework

3.2 Chromosome encoding and operators

3.3 Local search and heuristics (Should not appear here)

4 Preliminary result

4.1 Simulation data methodology

- Adopt the standard benchmark datasets: TR, AP and CAB
- Create a dataset from the map of a few provinces in the Central Region.

4.2 Experimental settings

4.3 Numerical Analysis

4.4 Results and findings

5 Conclusion and Future Work

- What did we do (and contribute?)
- What are the limitations?
- Future works

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